```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D5_Multi_Document_RAG> python w7d5_multi_document_rag.py

===========================================================================
W7D5 — MULTI-DOCUMENT RAG SYSTEM
===========================================================================
Technology: LlamaIndex + Ollama
LLM: llama3.2:3b
Embedding: nomic-embed-text:latest
Documents: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D5_Multi_Document_RAG\documents
Persistent index: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D5_Multi_Document_RAG\storage

===========================================================================
STEP 1 — VERIFYING DOCUMENTS
===========================================================================
Document directory: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D5_Multi_Document_RAG\documents
Number of documents: 5
  √ artificial_intelligence.txt
  √ cloud_computing.txt
  √ Document 1.txt
  √ software_design.txt
  √ software_engineering.txt

===========================================================================
STEP 2 — CONFIGURING LOCAL OLLAMA
===========================================================================
LLM model: llama3.2:3b
Embedding model: nomic-embed-text:latest
Ollama URL: http://localhost:11434

√ Ollama LLM configured
√ Ollama embedding model configured
√ SentenceSplitter configured
Chunk size: 1024
Chunk overlap: 100
```

ENG IN 15:40 24-08-2026

```
LLM timeout: 600.0 seconds

===========================================================================
STEP 3 — LOADING DOCUMENTS
===========================================================================
Loaded document objects: 5

===========================================================================
STEP 4 — VECTOR INDEX
===========================================================================
No usable persistent index found.
Creating a new vector index...

Embedding documents using Ollama.
This may take time on the first run.
Applying transformations: 100%|██████████| 1/1 [00:29<00:00, 29.06s/it]
Generating embeddings: 100%|██████████| 2048/2048 [18:02:11<00:00, 31.70s/it]
Generating embeddings: 100%|██████████| 1689/1689 [36:44<00:00,  1.31s/it]

√ Vector index created.
Indexing time: 67187.93 seconds
Index persisted to: C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D5_Multi_Document_RAG\storage

===========================================================================
STEP 5 — CREATING QUERY ENGINE
===========================================================================
√ Query engine created
Top-K retrieved chunks: 2
Response mode: simple_summarize
LLM: llama3.2:3b

===========================================================================
RAG QUERY
===========================================================================
Question:
What is artificial intelligence?
```

ENG IN 15:40 24-08-2026

```
ANSWER
--------------------------------------------------------------------------------
A field of research that focuses on creating machines that can perform tasks that typically require human intelligence, such as learn
ing, problem-solving, and decision-making.

SOURCES
--------------------------------------------------------------------------------
  - artificial_intelligence.txt (score=0.5440)

Query time: 71.31 seconds

================================================================================
STEP 6 — RUNNING 10 TEST QUESTIONS
================================================================================

################################################################################
QUESTION 1/10
################################################################################

================================================================================
RAG QUERY
================================================================================
Question:
What is artificial intelligence and what are its major applications?

ANSWER
--------------------------------------------------------------------------------
Artificial intelligence (AI) refers to the development of computer systems that can perform tasks that typically require human intell
igence, such as learning, problem-solving, and decision-making.

AI has numerous applications across various industries, including but not limited to:

1. Virtual assistants: AI-powered virtual assistants, like chatbots, help users with tasks, answer questions, and provide information
.
2. Image and speech recognition: AI algorithms can recognize and interpret images and speech patterns, enabling applications such as
facial recognition, voice assistants, and self-driving cars.
```

```
3. Healthcare: AI is used in medical diagnosis, personalized medicine, and predictive analytics to improve patient outcomes and strea
mline clinical workflows.
4. Finance: AI is applied in risk management, portfolio optimization, and automated trading to enhance financial decision-making.
5. Education: AI-powered adaptive learning systems personalize education, provide real-time feedback, and automate grading.
6. Customer service: AI-driven chatbots and virtual assistants help customers with queries, provide support, and route complex issues
 to human representatives.
7. Cybersecurity: AI-powered systems detect and respond to cyber threats, predict vulnerabilities, and improve incident response.
8. Autonomous vehicles: AI enables self-driving cars to navigate roads, detect obstacles, and make decisions in real-time.
9. Natural language processing: AI algorithms analyze and understand human language, enabling applications such as language translati
on, sentiment analysis, and text summarization.
10. Robotics: AI is used in robotics to enable machines to perceive their environment, make decisions, and interact with humans.

These are just a few examples of the many applications of artificial intelligence. As AI continues to evolve, we can expect to see ev
en more innovative uses of this technology in the future.

SOURCES
--------------------------------------------------------------------------------
  - artificial_intelligence.txt (score=0.5710)

Query time: 38.65 seconds

################################################################################
QUESTION 2/10
################################################################################

================================================================================
RAG QUERY
================================================================================
Question:
What are the main characteristics of cloud computing?

ANSWER
--------------------------------------------------------------------------------
Cloud computing is a model of delivering computing services over the internet, where resources such as servers, storage, databases, s
oftware, and applications are provided as a service to users on-demand. The main characteristics of cloud computing include:

1. **On-demand self-service**: Users can provision and de-provision resources as needed, without requiring human intervention.
```

```
1. **On-demand self-service**: Users can provision and de-provision resources as needed, without requiring human intervention.
2. **Broad network access**: Resources are accessible over the internet, or a private network, from any device, anywhere in the world
.
3. **Resource pooling**: Resources are pooled together to provide a multi-tenant environment, where resources can be dynamically allo
cated and re-allocated based on demand.
4. **Rapid elasticity**: Resources can be quickly scaled up or down to match changing business needs.
5. **Measured service**: Users only pay for the resources they use, rather than having to purchase and maintain their own hardware an
d software.
6. **Multi-tenancy**: Resources are shared among multiple users, with each user's resources being isolated from those of other users.
7. **Scalability**: Cloud computing resources can be scaled up or down to match changing business needs.
8. **Reliability**: Cloud computing providers typically offer high levels of uptime and redundancy, with built-in redundancy and fail
over capabilities.
9. **Security**: Cloud computing providers typically offer a range of security features, including data encryption, access controls,
and monitoring.
10. **Flexibility**: Cloud computing resources can be easily moved between different locations, or between different cloud providers,
 as needed.

SOURCES
---------------------------------------------------------------------------
  - cloud_computing.txt (score=0.5538)

Query time: 97.62 seconds

###########################################################################
QUESTION 3/10
###########################################################################

===========================================================================
RAG QUERY
===========================================================================
Question:
What are the important principles of software design?

ANSWER
---------------------------------------------------------------------------
Modularity, scalability, maintainability, and usability are key considerations in software design.
```

```
SOURCES
---------------------------------------------------------------------------
  - software_engineering.txt (score=0.4964)
  - software_design.txt (score=0.4889)

Query time: 33.74 seconds

###########################################################################
QUESTION 4/10
###########################################################################

===========================================================================
RAG QUERY
===========================================================================
Question:
What are the major stages involved in software engineering?

ANSWER
---------------------------------------------------------------------------
Design, Implementation, Testing, and Maintenance are the major stages involved in software engineering.

SOURCES
---------------------------------------------------------------------------
  - software_engineering.txt (score=0.5406)

Query time: 18.20 seconds

###########################################################################
QUESTION 5/10
###########################################################################

===========================================================================
RAG QUERY
===========================================================================
Question:
How does software architecture influence software design?
```

Windows PowerShell

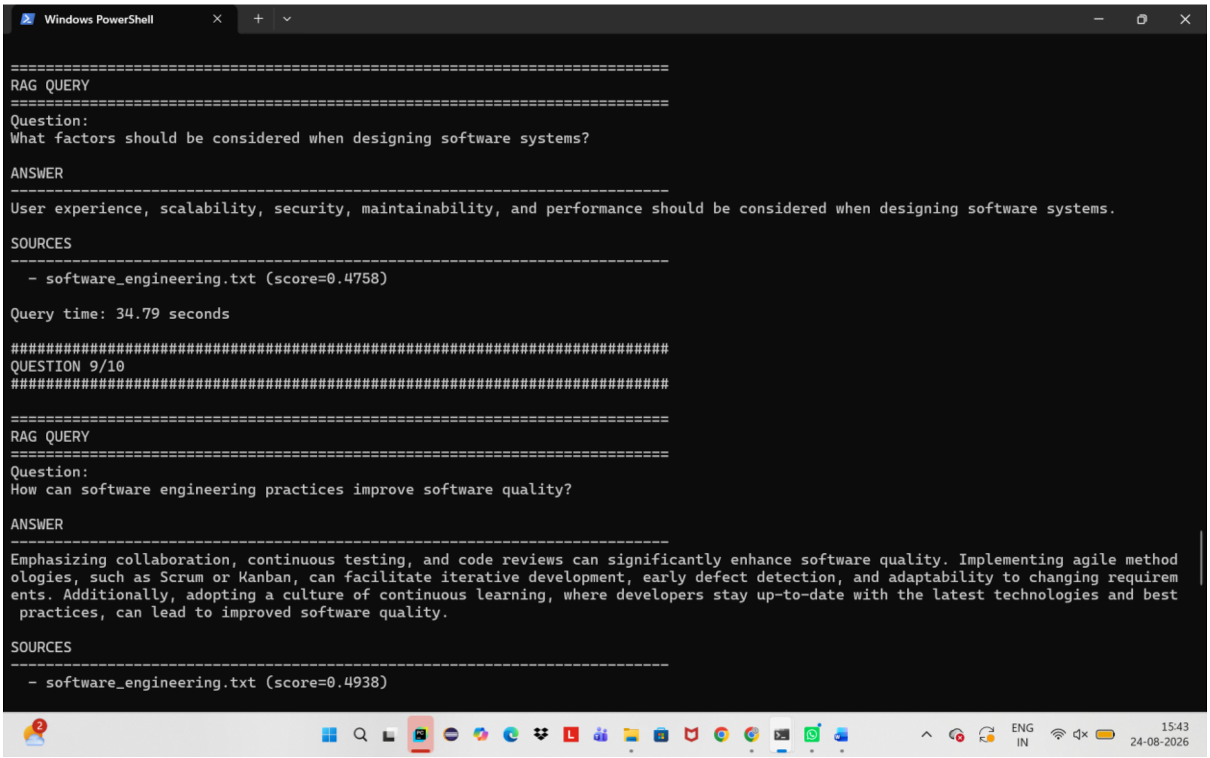
```
ANSWER
------------------------------------------------------------------------------
The relationship between software architecture and design is multifaceted. Software architecture serves as the foundation for softwar
e design, providing a blueprint for the overall structure and organization of the system. It defines the overall pattern and structur
e of the system, including the relationships between different components, and the principles that guide their interactions.

Influenced by software architecture, software design focuses on the detailed implementation of the system, including the selection of
 specific technologies, algorithms, and data structures. The design must be aligned with the overall goals and constraints of the sof
tware architecture, ensuring that the system meets its functional and performance requirements.

Effective software design that is influenced by software architecture typically involves:

1. **Modularity**: Breaking down the system into smaller, independent components that can be developed, tested, and maintained separa
tely.
2. **Abstraction**: Focusing on the essential features and behaviors of the system, hiding non-essential details and complexities.
3. **Reusability**: Designing components and modules that can be reused across multiple applications and contexts.
4. **Scalability**: Ensuring that the system can adapt to changing requirements and growth in complexity.
5. **Flexibility**: Allowing for easy modification and adaptation to changing circumstances.

By aligning software design with software architecture, developers can create systems that are more maintainable, efficient, and effe
ctive in meeting the needs of users.

SOURCES
------------------------------------------------------------------------------
  - software_engineering.txt (score=0.4754)

Query time: 133.83 seconds

##############################################################################
QUESTION 6/10
##############################################################################

==============================================================================
RAG QUERY
==============================================================================
Question:
```

ENG IN 15:42 24-08-2026

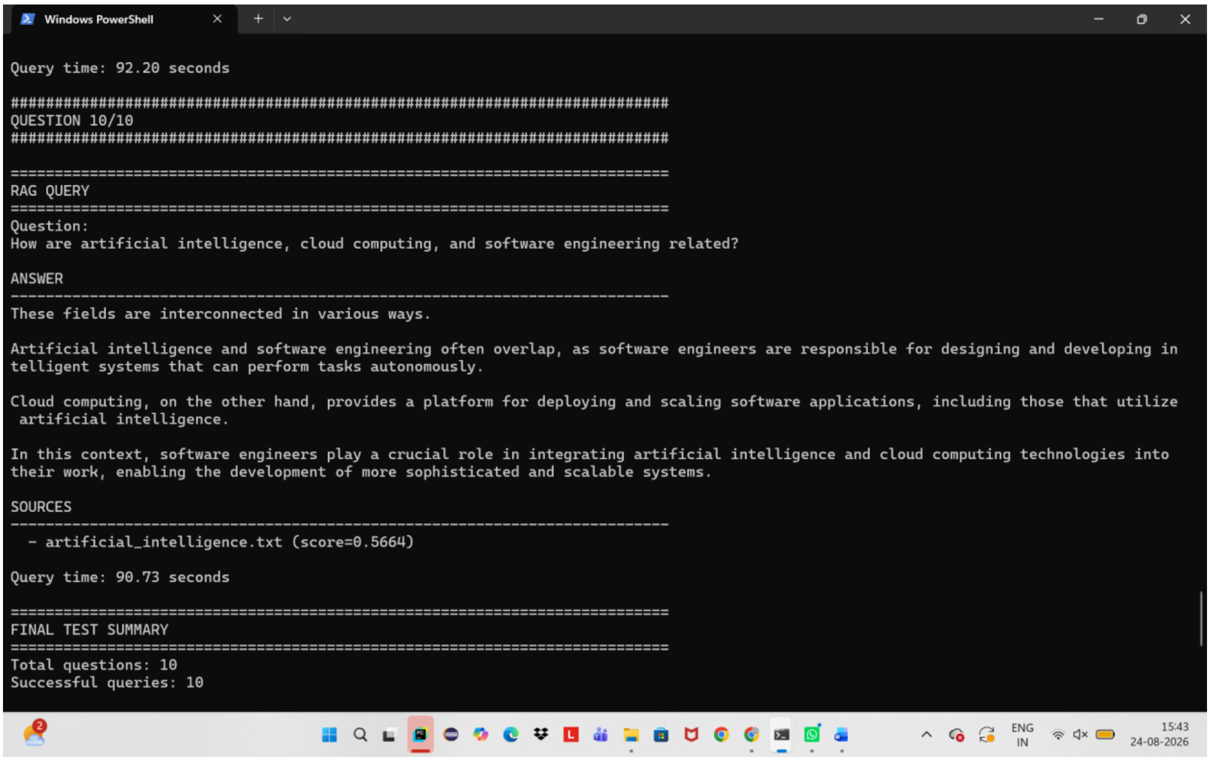
Windows PowerShell

```
What are the advantages of using cloud computing?

ANSWER
------------------------------------------------------------------------------
Increased scalability, reduced capital expenditures, improved resource utilization, enhanced collaboration and accessibility, and fas
ter deployment of applications are some of the key benefits of cloud computing.

SOURCES
------------------------------------------------------------------------------
  - cloud_computing.txt (score=0.5221)

Query time: 11.85 seconds

##############################################################################
QUESTION 7/10
##############################################################################

==============================================================================
RAG QUERY
==============================================================================
Question:
What are common techniques used in artificial intelligence?

ANSWER
------------------------------------------------------------------------------
Machine learning, deep learning, natural language processing, and computer vision are often employed in artificial intelligence.

SOURCES
------------------------------------------------------------------------------
  - artificial_intelligence.txt (score=0.6011)

Query time: 45.55 seconds

##############################################################################
QUESTION 8/10
##############################################################################
```

ENG IN 15:43 24-08-2026



```
===========================================================================
RAG QUERY
===========================================================================
Question:
What factors should be considered when designing software systems?

ANSWER
---------------------------------------------------------------------------
User experience, scalability, security, maintainability, and performance should be considered when designing software systems.

SOURCES
---------------------------------------------------------------------------
  - software_engineering.txt (score=0.4758)

Query time: 34.79 seconds

###########################################################################
QUESTION 9/10
###########################################################################

===========================================================================
RAG QUERY
===========================================================================
Question:
How can software engineering practices improve software quality?

ANSWER
---------------------------------------------------------------------------
Emphasizing collaboration, continuous testing, and code reviews can significantly enhance software quality. Implementing agile method
ologies, such as Scrum or Kanban, can facilitate iterative development, early defect detection, and adaptability to changing requirem
ents. Additionally, adopting a culture of continuous learning, where developers stay up-to-date with the latest technologies and best
 practices, can lead to improved software quality.

SOURCES
---------------------------------------------------------------------------
  - software_engineering.txt (score=0.4938)
```

```
Query time: 92.20 seconds

###########################################################################
QUESTION 10/10
###########################################################################

===========================================================================
RAG QUERY
===========================================================================
Question:
How are artificial intelligence, cloud computing, and software engineering related?

ANSWER
---------------------------------------------------------------------------
These fields are interconnected in various ways.

Artificial intelligence and software engineering often overlap, as software engineers are responsible for designing and developing in
telligent systems that can perform tasks autonomously.

Cloud computing, on the other hand, provides a platform for deploying and scaling software applications, including those that utilize
 artificial intelligence.

In this context, software engineers play a crucial role in integrating artificial intelligence and cloud computing technologies into
their work, enabling the development of more sophisticated and scalable systems.

SOURCES
---------------------------------------------------------------------------
  - artificial_intelligence.txt (score=0.5664)

Query time: 90.73 seconds

===========================================================================
FINAL TEST SUMMARY
===========================================================================
Total questions: 10
Successful queries: 10
```

```
Query time: 90.73 seconds

===========================================================================
FINAL TEST SUMMARY
===========================================================================
Total questions: 10
Successful queries: 10
Failed queries: 0
Average successful query time: 59.72 seconds

Individual query times:
  Query 1: 38.65 seconds
  Query 2: 97.62 seconds
  Query 3: 33.74 seconds
  Query 4: 18.20 seconds
  Query 5: 133.83 seconds
  Query 6: 11.85 seconds
  Query 7: 45.55 seconds
  Query 8: 34.79 seconds
  Query 9: 92.20 seconds
  Query 10: 90.73 seconds

✓ ALL 10 TEST QUESTIONS PASSED

===========================================================================
W7D5 MULTI-DOCUMENT RAG COMPLETED
===========================================================================
✓ Multiple documents loaded
✓ Ollama embeddings configured
✓ Local Ollama LLM configured
✓ LlamaIndex vector index created/loaded
✓ QueryEngine created
✓ Sources displayed
✓ 10 test questions executed
✓ Index persisted for future runs
PS C:\Users\Sanjana CA\OneDrive\Documents\CynarisInternship\W7D5_Multi_Document_RAG>
```